

# LECTURE 1

## INTRODUCTION TO DATABASES. DATA MODELS.

### RELATIONAL DATABASES. DBMS

**Objective:** To develop a basic understanding of database concepts, introduce students to various data models, and provide fundamental knowledge of relational databases and database management systems (DBMS).

#### Key Topics

1. The concept and purpose of databases
2. The evolution of data models: hierarchical, network, and relational
3. Key concepts of the relational data model
4. Components and functions of DBMS
5. Advantages and disadvantages of different approaches to data organization

#### Introduction to Databases

A database (DB) is an organized collection of structured data stored in the memory of a computer system and displaying the state of objects and their relationships in the subject area under consideration.

#### *Main characteristics of databases:*

- Permanence of data storage;
- Structure and organization;
- Independence from application programs;
- Multi-user access with concurrency control;
- Protection from unauthorized access;
- Ensuring data integrity and recovery.

#### Aspects of Database Classification

Databases are usually classified according to the following aspects.

##### *1. According to the form of information presentation:*

###### *1.1. text:*

- structured;
- not structured;
- partially structured;

###### *1.2. video;*

###### *1.3. audio;*

###### *1.4. multimedia.*

##### *2. By data model type:*

###### *2.1. Network;*

###### *2.2. Hierarchical;*

###### *2.3. Relational;*

###### *2.4. Post-relational (multidimensional - Cache, Teradata);*

*2.5. Object-oriented (Cache, FastObjects, GemStone/S, Jasmine, ObjectStore, Objectivity/DB, Versant).*

##### *3. By type of stored information:*

###### *3.1. Documentary:*

- Bibliographic;
- Documentary;
- Full-text;

- 3.2. Factual;
- 3.3. Lexicographic:
  - Dictionaries;
  - Classifiers.

4. *By the nature of data organization and access method:*

- 4.1. Personal (local);
- 4.2. General (integrated):
  - Centralized;
  - Distributed.

5. *By subject area:*

- 5.1. Business information sector (stock exchange, financial, commercial);
- 5.2. Professional information sector (patent, legal, technological);
- 5.3. Mass consumer information sector (directories, schedules).

6. *By rate of change and data storage time:*

- 6.1. Operational (transactional);
- 6.2. Historical (data warehouses).

## **HISTORICAL EVOLUTION OF DATA MODELS**

### ***Hierarchical Data Model***

The hierarchical model organizes data into a tree structure, where each node has only one ancestor. This model was popular in the 1960s and 1970s.

*Key Features:*

- Strict tree structure
- One-to-many relationships
- Fast access to data along a known path
- Difficulty changing the structure
- Example: IMS (Information Management System) from IBM

*Disadvantages:*

- Difficulty representing complex relationships
- Data redundancy
- Structural rigidity

### ***Network Data Model***

The network model extends the hierarchical model by allowing nodes to have multiple ancestors. Developed as part of the CODASYL conference.

*Key Features:*

- Graph data structure
- Many-to-many relationships
- More flexible than a hierarchical model
- Complex to design and use
- Example: IDMS (Integrated Database Management System)

### ***The Relational Data Model***

Proposed by Edgar Codd in 1970 in his paper "A Relational Model of Data for Large Shared Data Banks," it became the dominant model for decades to come.

*Key Concepts:*

- Data is represented as tables (relations)
- Rows (tuples) and columns (attributes)

- Mathematical foundation - set theory and relational algebra
- Declarative query language (SQL)

A relation is a fundamental concept, representing a table with the following properties:

- Each row is unique;
- The order of rows is undefined;
- All values in a column are homogeneous;
- Column names are unique within the table;

A relation schema is a description of the structure of a relation, including:

- Attribute names;
- Attribute data types;
- Integrity constraints;

Keys:

- Primary key - a unique identifier for a tuple;
- Foreign key - an attribute referencing the primary key of another relation;

Advantages of the relational model:

- Simplicity and mathematical soundness
- Data independence
- Flexibility when changing structure
- Powerful SQL query language
- Widespread adoption and support

## **DATABASE MANAGEMENT SYSTEMS (DBMS)**

A DBMS is a software suite designed to create, maintain, and use databases.

### ***Key DBMS Functions***

Data Management:

- Creating, modifying, and deleting data structures;
- Data manipulation (insert, update, delete);
- Query execution optimization;

Transaction Management:

- ACID Properties:
  - Atomicity - a transaction is executed in its entirety or not at all;
  - Consistency - a transaction transitions the database from one consistent state to another;
  - Isolation - concurrent transactions do not interfere with each other;
  - Durability - the results of a completed transaction are preserved after failures.

### ***Concurrency Control:***

- Locking Mechanisms;
- Concurrency Control Protocols;
- Conflict resolution.

Disaster recovery:

- Change logging;
- Backup;
- Recovery from hardware and software failures.

### **DBMS Architecture**

Three-tier ANSI/SPARC architecture:

1. Internal layer - physical representation of data on the storage medium

2. Conceptual layer - logical structure of the entire database
3. External layer - data representation for specific users

### **Modern Trends**

NoSQL databases - an alternative to relational databases for specific tasks:

- Document-oriented (MongoDB)
- Key-value (Redis)
- Column-oriented (Cassandra)
- Graph (Neo4j)

NewSQL - systems that combine the scalability of NoSQL and the ACID guarantees of relational DBMSs.

### **Quiz Questions**

1. What is a database and what are its main characteristics?
2. What are the main differences between the hierarchical, network, and relational data models?
3. List the main concepts of the relational data model.
4. What are the main functions of a DBMS?
5. What are the ACID properties of transactions? Explain each of them.
6. What advantages of the relational model have contributed to its widespread adoption?

### **References**

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3. Kuznetsov, S.D. Database Fundamentals. 2nd ed. Chapters 1-2.
4. Khomenko A.D., Tsygankov V.M., Maltsev M.G. Databases. - Chapter 1.